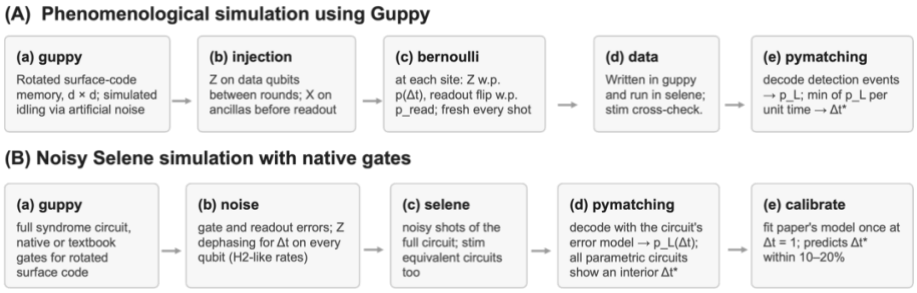
Optimal syndrome timing on trapped-ion hardware

Problem statement

Surface-code memories usually run syndrome extraction back-to-back. Measuring less often lets idle errors build up and measuring more often adds gate and readout faults. Thus, there exists a trade-off between idling and syndrome extraction leading to an optimal interval Δt^* that minimises logical error per unit time [1]. We test whether the optimal interval thesis is valid with native-gate, gate errors and coherent idle noise in a trapped-ion device. We go a step further in validating if it holds on Quantinuum hardware, by assuming a Z-biased dephasing noise, and later designing and executing experiments on H1/H2 emulators provided through Aqora.

Approach

To test that the idling-noisy syndrome extraction trade-off holds in principle, we run two simulation pipelines described below:



Simulation pipelines.
(a) Phenomenological simulation through noise injection using Guppy.
(b) Noisy Selene simulation with noisy native gates, and dephasing.

Symbols.
- d: code distance
- Δt: simulated interval between syndrome extractions,
- $p(\Delta t)$: Z-error probability per data qubit for Δt
- p_{read}: readout-flip probability
- Native Gates: the hardware's own gate set, as opposed to ideal gates
- $p_{\text{data}}, p_{\text{read}}$: the data and readout error rates fitted to paper, measured once from detection events at $\Delta t = 1$
- p_L: logical error probability
- Δt^*: interval that minimises logical error per unit time.

After simulations, while approach (A) is implemented using Guppy-> QIR bridge on Aqora hosted H2 emulators, for (B) additional consideration exists. In (B), we model Quantinuum like hardware using dephasing-Z to each data qubit over the simulated interval Δt under one of two idle models: Markovian at incoherent rate γ, and/or coherent phase drift at rate c, as follows:

$$p_{\text{Markov}}(\Delta t) = 1/2\,(1 - e^{-\gamma \Delta t}), \qquad p_{\text{coherent}}(\Delta t) = sin^2(c\Delta t/2).$$

Rather than directly implementing approach 2. on emulators, we design experiments to extract **γ and c** using spectator qubits.

Key results

1. The optimum survives full circuit noise. In Fig. 1 (native gates; idle models linear, $\gamma = 0.3$, quadratic, c = 0.01 and mixed), the logical error per unit time falls and then rises for all $d = 3, 5, 7$. This corresponds to approach (B). tested on Selene with native gates and circuit noise.

2. A phenomenological model predicts it from one calibration. Calibrated once from detection events at $\Delta t = 1$, the model introduced in the paper predicts Δt^* within ±15–20% at $d = 3$, and ±10% at $d = 5, 7$ (solid lines in Fig. 1).

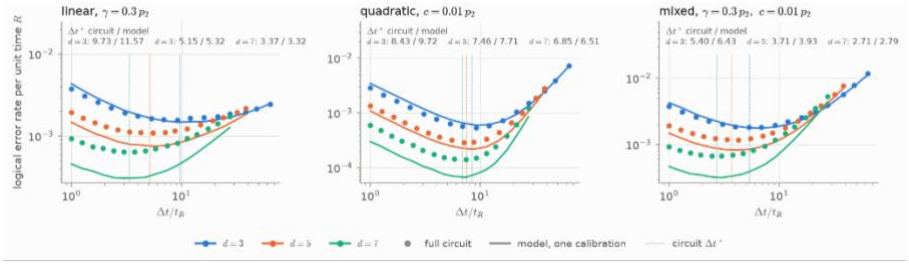


Figure 1: Simulations for approach 2. assuming Quantinuum-device like idling noise, and gate errors applied on data as well as ancillae.The left panel is for linear idling, the second for quadratic and third for a mixture of the two. Non trivial , optimal syndrome interval is observed in all the cases.

3. Expected tradeoff on H2. Fig. 2 maps the optimal interval $\Delta t^*/t_R$ over the idle plane at $d = 3,5,7$: linear idle strength γ/p_2 against quadratic idle strength c/p_2, both relative to the two-qubit gate error p_2. Beyond the white contour, waiting gains less than 5% over $\Delta t = 1$. The balck line places H2-1E idle rates by round duration t_R; it stays inside the gain region up to $t_R \approx$ 100–300 ms, where it crosses the white line.

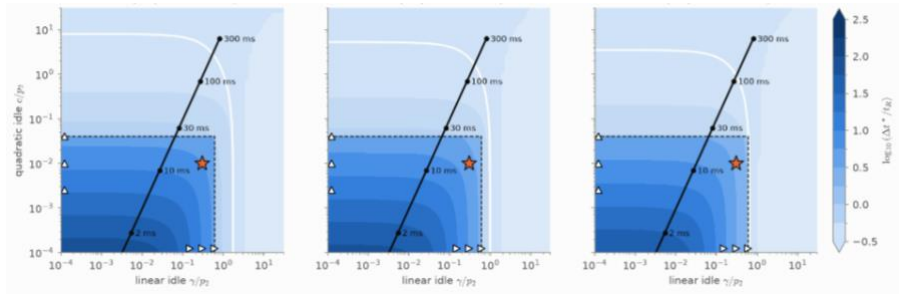


Figure 2: Heatmap for optimal Δt as a function of idling parameters at different distances across the panels. Darker region represents the possibility of optimizing the interval with boundary of feasible region marked by white contour. Black line =*

4. Experiments on H2 emulators. Guppy programs, compiled to QIR, ran on H2-1E/H2-2E through aqora.
(a) *Approach (A) works:* a d = 5 memory with in-program Z noise matches Stim within large error bars, see Fig 3.
(b) *The optimum survives device noise:* a d = 5 Δt sweep on H2-2E follows a gate-level model of the program with H2-2E reported gate errors within ~1σ. The U-curve is higher and flatter but keeps its interior minimum.
(c) *Idle parameters:* anchored spectator qubits show no consistent dephasing up to 128 layers (< 0.5%). γ and c remain unresolved because longer waits exceed the per-job cost limit, and physical t_R is not reported by Quantinuum on spec sheet. (plots and results not shown here for brevity)

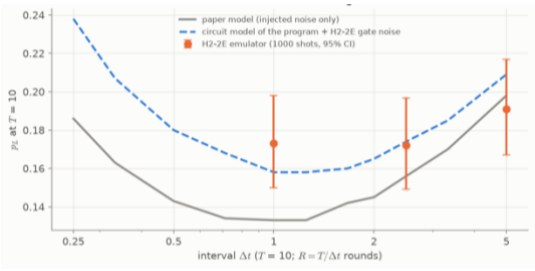


Figure 3: Rotated surface code with X injected noise emulating idling, tested on H2 emulator.

Potential impact

Algorithms leave most logical qubits idle: sequential workloads spend 84–87% (adder) and 41–51% (QFT) of their qubit-time waiting, so how an idle memory is protected governs most of a computation's space-time volume.

Time can stand in for space. For idling qubits a longer syndrome interval lowers logical error per unit time: waiting at distance d can beat back-to-back extraction at d + 2, saving 16–24 data qubits per logical qubit at short round durations (shown in Fig 01 and 09 in /figures in the repo).

Generalised across noise models. $\Delta t^*$is mapped over linear and quadratic idle noise (γ, c) with gate noise, not one fixed phenomenological model. The access to machine with deeper circuits and more shots can reveal whether an optimal syndrome extraction interval indeed exists.

Linked to hardware feasibility. Placing H2-1E idle rates on that map ties the gain to the round duration: waiting helps up to $t_R \approx$ 100–300 ms, and one calibration at $\Delta t = 1$ predicts Δt^* within 10–20%.

Reference. T. Haug, K. Bharti, L. Aolita, *Exponential logical-error reduction in quantum memories via optimal syndrome-measurement timing*, arXiv:2608.06242 (2026).